

Anirban Basu

RESEARCHER IN COMPUTATIONAL TRUST, PRIVACY, SECURITY AND ARTIFICIAL INTELLIGENCE

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"There is only one corner of the universe you can be certain of improving, and that's your own self." – Aldous Huxley

Summary

With more than a decade and a half of research experience within both academia and industry, I have co-authored 85+ research papers and 30+ Japanese patents. My publications have been cited more than 3,100 times setting my overall h-index to 25 (source: Google Scholar). My current research is based on a user-empowering view of interactions between humans and artificial intelligence; and the role of computational trust in security and privacy. I like working on complex problems that require multi-disciplinary approaches. Outside research and experimental stuff (most of which, I make open-source), I am keen on photography, videography and the associated travels.

Professional experience

Hitachi, Ltd.

Yokohama, Japan

SENIOR RESEARCHER (APR. 2021 – PRESENT) AND RESEARCHER (OCT. 2018 – MAR. 2021)

Oct. 2018 – present

- **Summary:** Researcher, working on several research projects on computational trust, privacy and security for artificial intelligence within the Hitachi Research & Development group. Also taught about trust through Hitachi Academy between Apr. 2022 – Mar. 2023.

Freelancer

Tokyo, Japan

CONSULTANT

Aug. 2014 – present

- **Highlights:** Open-source software development, web design and development, document design and photography.

KDDI Research

Fujimino, Japan

RESEARCH ENGINEER

May. 2013 – Apr. 2018

- **Highlights:** Worked on a number of private and government funded projects on privacy, trust and security. Generated 30+ research publications and around 20 patents.

Tokai University

Tokyo, Japan

POST-DOCTORAL RESEARCHER

Oct. 2010 – Mar. 2013

- **Highlights:** Worked on a multi-institutional government funded JPY 500M+ project on privacy-preserving collaborative filtering.

University of Sussex

Brighton, UK

ASSOCIATE TUTOR (OCT. 2004 – JAN. 2010) AND RESEARCH FELLOW (OCT. 2008 – SEP. 2009)

Oct. 2004 – Jan. 2010

- **Highlights:** Re-developed a course module on electronic commerce technologies for the Masters degree programme funded by American Express. Worked in two (GBP 1M and GBP 1.4M) EPSRC funded research projects related to computational trust.

Freelancer

Brighton, UK

TECHNOLOGY CONSULTANT (JUN. 2010 – SEP. 2010; JUL. 2005 – SEP. 2008)

Jul. 2005 – Sep. 2010

- **Highlights:** Technology consultant to the University of Sussex library. Developed Java Web Services hooks for Class Calendar Ltd. Co-developed computational skills training videos for Claro Training Ltd.

University of Sussex

Brighton, UK

RESEARCH ASSISTANT (SEP. 2003 – SEP. 2004) AND STEP UK SUMMER INTERN (JUL. 2003 – AUG. 2003)

Jul. 2003 – Sep. 2004

- **Highlights:** Developed a lightweight augmented reality based visualisation technology based on backend XML-based storage as part of a EUR 2.8M EU Framework Programme V funded project on digital cultural heritage.
- **Award:** Shell Technology Enterprise Programme (STEP, UK) Local Finalist.

Professional affiliations

University of Sussex

Brighton, UK

HONORARY RESEARCH FELLOW (MAR. 2020 – PRESENT) AND VISITING RESEARCH FELLOW (JAN. 2010 – MAR. 2020)

Jan. 2010 – present

- Collaborative research with the Foundations of Software Systems (FoSS) group in the Department of Informatics.

IFIP Trust Management (IFIPTM)

Tokyo, Japan

SECRETARY (JUN. 2015 – PRESENT) AND PUBLICITY CHAIR (JUN. 2010 – JUN. 2015)

Jun. 2010 – present

- Contribution to the IFIP Working Group 11.11 – Trust Management (IFIPTM), a well-known venue on computational trust management research.

OntarioTech University

Oshawa, Canada

ADJUNCT PROFESSOR

Oct. 2022 – Sep. 2025

- Collaborative research with Dr. Peter Lewis and Dr. Stephen Marsh within the Faculty of Business and Information Technology.

Rutgers University

Newark, NJ, USA

VISITING RESEARCH FELLOW

Oct. 2014 – Sep. 2017

- Collaborative research with Prof. Jaideep Vaidya at the Rutgers Business School.

Education

University of Sussex

DOCTOR OF PHILOSOPHY (D.PHIL.) IN COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Brighton, UK

Oct. 2004 – Feb. 2010

- **Supervisors:** Prof. Ian Wakeman and Dr. Dan Chalmers.
- **Examiners:** Dr. Des Watson (University of Sussex) and Prof. David Parish (Loughborough University)
- **Doctoral thesis:** 'A Reputation Framework for Behavioural History' proposed a high-level, generalised, policy-independent and context-aware framework to build and share reputations of network clients based on their behavioural history in order to inform future interactions. PDF: <https://www.anirbanbasu.com/files/thesis-dphil2010.pdf>.

University of Sussex

BACHELOR OF ENGINEERING (B.ENG.) IN COMPUTER SYSTEMS ENGINEERING

Brighton, UK

Oct. 2001 – Jun. 2004

- **Grade:** First Class Honours.
- **Awards:** (1) IEE Sussex Branch local prize for Outstanding Performance in Computer Systems Engineering (B.Eng.); and (2) the Eurotherm company prize for Best Electrical, Electronic or Computer Systems Engineering project.
- **Final year project:** 'ARCOLite – a visualisation system for viewing museum artefacts' presented a lightweight XML-based repository and a desktop visualisation in both virtual and augmented reality. PDF: <https://www.anirbanbasu.com/files/thesis-beng2004.pdf>.

Accolades

INVITED TALKS

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| 2018 | World Wide Web Foundation , Privacy – connecting the dots | London, UK |
| 2017 | IFIP TM – Doctoral Symposium , Homomorphic Encryption – an answer to privacy? | Göteborg, Sweden |
| 2015 | Kyushu University – Next Generation Cryptography , Homomorphic Encryption – are we there yet? | Fukuoka, Japan |
| 2012 | CSA – SecureCloud , Practical privacy using homomorphic encryption – a myth or reality | Frankfurt, Germany |

AWARDS

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|------|---|--------------|
| 2004 | IEE Sussex Branch local prize , for Outstanding Performance in Computer Systems Engineering (B.Eng.) | Brighton, UK |
| 2004 | Eurotherm company prize , for Best Electrical, Electronic or Computer Systems Engineering project | Brighton, UK |
| 2003 | Local Finalist , Shell Technology Enterprise Programme (STEP) | Brighton, UK |

SCHOLARSHIPS

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|------|---|--------------|
| 2004 | Graduate Teaching Assistantship (2004–2007) , Department of Informatics (University of Sussex) | Brighton, UK |
| 2001 | Dean's Bursary (2001–2004) , Department of Engineering and Design (University of Sussex) | Brighton, UK |

Publications

A reverse chronological list of my publications is available at: <https://www.anirbanbasu.com/files/publications-by-year.pdf>.

Research papers

SELECTED PUBLICATIONS

- [1] **A. Basu**, Y. Kaga, and K. Fujiwara, "A framework for the estimation of policy-based trustworthiness of AI agents," in *To appear in the proceedings of the 23rd Annual International Conference on Privacy, Security, and Trust*. Ottawa, Canada: IEEE, August 2026.
- [2] **A. Basu** and M. Yoshino, "Confidential data anonymisation using trusted execution environments," *To appear in the IEEE Access*, 2026.
- [3] **A. Basu**, M. Yoshino, and M. Toba, "Practical confidential data cleaning using trusted execution environments," in *Proceedings of the IEEE International Conference on Cyber Security and Resilience (CSR)*, Chania, Greece, August 2025, pp. 342–349.
- [4] **A. Basu** and S. Kai, "Software and its perceived trustworthiness," in *Proceedings of the IFIP WG 11.11 International Conference on Trust Management (IFIPTM)*. Amsterdam, Netherlands: Springer Nature, Switzerland, October 2023, pp. 105–120.
- [5] S. Marsh, T. Atele-Williams, **A. Basu**, N. Dwyer, P. R. Lewis, H. Miller-Bakewell, and J. Pitt, "Thinking about Trust: People, Process, and Place," *Elsevier Cell Patterns*, vol. 1, June 2020.

Patents and public information (total 36)

WIPO PATENTSCOPE PUBLICATION NUMBERS

2024068732, 2024035496, WO/2024/257315, 2022163431, 2022102062, 2021170262, 2021089679, WO/2021/112216, 2020014056, 2019176264, 2019174995, 2019121946, 2019075619, 2018195907, 2018195154, 2018136625, 2018136626, 2018136627, 2018124924, 2018124513, 2018022258, 2017183932, 2017059872, 2017037180, 2016167705, 2016052047, 2016025563, 2016009243, 2015210795, 2015176496, 2015148907, 2015127859, 2015095671, 2015069350, 2015041863, 2015014958.

Skills

Computing skills

OS: macOS, Linux, Windows; *Documents*: MS Office, Google Docs and $\LaTeX$; *Software development*: Python, Golang (Go), Rust, Java, C#, Objective-C, C++, C, Visual Basic, XML, Javascript, UML, Git. (**Non-exhaustive list.**)

Languages

English (fluent), Bengali (native), Japanese (intermediate), Hindi (elementary)